

CyberEval: Role-Aware Multi-Dimensional Evaluation of LLM Security Competence

Anonymous Authors

Anonymous Institution

Abstract. LLMs are increasingly proposed for vulnerability triage, threat-intelligence analysis, secure code review, incident response, and governance support, but current cybersecurity benchmarks still emphasize narrow slices of that workflow space such as secure coding, offensive CTF solving, or cyber threat intelligence in isolation [4,5,6,7,8,9,10,11]. We present CYBEREVAL, a reproducible evaluation framework organized around seven security dimensions: Vulnerability Knowledge, Threat Intelligence, Secure Coding, Incident Response, Compliance & Governance, Forensic Analysis, and Security Architecture. The current public artifact contains a 210-item seed bank with deterministic difficulty annotations, a paired MCQ/scenario simulation protocol, role-weighted deployment scoring, and auditable response traces. In a deterministic simulation over 10 model profiles calibrated from public benchmark trends, Claude-3.5-Sonnet achieves the highest aggregate score at 60.5% (95% Wilson CI: 55.7–65.0), followed by GPT-4o at 56.9% (52.1–61.6). Across models, Secure Coding is the strongest dimension (mean 58.5%), while Security Architecture (22.7%) and Compliance & Governance (31.3%) remain persistent weak points. The average MCQ-to-scenario gap is 11.2 percentage points, indicating that MCQ-only evaluation materially overestimates deployment readiness. We emphasize that the current results are *not* live API evaluations: they come from a transparent simulation harness intended to stress-test benchmark structure, reporting, and role-aware analysis before expensive online execution. This positioning makes CYBEREVAL a framework paper and reproducible artifact rather than a finished large-scale leaderboard.

Keywords: LLM Security Evaluation · Cybersecurity Benchmarking · Role-Aware Assessment · Simulation Study · Security Reasoning

1 Introduction

Recent work has shown both the promise and the risk of deploying LLMs in cybersecurity. Studies of coding assistants found that LLM-supported development can yield insecure code and can induce misplaced confidence in users [13,14]. In parallel, offensive-security systems such as PentestGPT and LLM-guided penetration-testing agents demonstrate that model competence is not limited to code completion but extends to multi-step cyber tasks [15,16]. A recent literature review further argues that the cyber-LLM landscape is expanding rapidly

across vulnerability analysis, threat intelligence, malware analysis, secure development, and red teaming [12]. This breadth creates an evaluation problem: organizations do not need a single “cyber score,” but rather an account of *which* security workflows a model can support and where the failure modes remain.

Existing benchmarks only partially address that need. CyberSecEval focuses on secure coding and cyberattack safety [4,5,6]. CyberMetric and CTIBench target cybersecurity knowledge and threat intelligence, respectively [7,8]. NYU CTF Bench and CyBench emphasize offensive or agentic cyber capability [9,10]. SecBench moves closest to broad coverage, but the field still lacks a lightweight, auditable framework that explicitly supports *role-aware* model selection across security workflows [11]. In practice, an application-security engineer, a SOC analyst, and a GRC reviewer care about very different mixtures of secure coding, incident response, governance, and architectural reasoning.

CYBEREVAL is designed as a reproducible first step toward that goal. Rather than presenting a new live leaderboard, we release a deterministic evaluation harness that makes three design choices explicit: (i) seven workflow-oriented security dimensions, (ii) a paired-format protocol that distinguishes easier direct-answer evaluation from harder scenario-weighted reasoning, and (iii) role-weighted aggregation and deficit scoring for deployment analysis. This framing is motivated by general LLM evaluation research showing that benchmark conclusions depend strongly on task design, format, and calibration methodology [1,2,3].

Our main contributions are:

- **A reproducible multi-dimensional cyber-evaluation artifact.** The released framework contains a 210-item seed bank spanning seven security dimensions, deterministic simulation seeds, generated figures, and auditable response traces.
- **Role-aware scoring and risk analysis.** In addition to aggregate scores, CYBEREVAL computes workflow-specific suitability scores and deficit-based deployment risk for four illustrative roles: SOC analyst, application-security engineer, GRC analyst, and security architect.
- **An honest simulation study with uncertainty estimates.** We report Wilson confidence intervals, difficulty-skew diagnostics, and format-gap analysis, while explicitly documenting that the current release is a benchmark-construction prototype rather than a live model-comparison benchmark.

The most important empirical result is qualitative rather than leaderboard-centric: different security dimensions remain sharply uneven even among the strongest profiles, and the paired-format protocol exposes an 11.2-point performance drop from MCQ-style answers to harder scenario-weighted evaluation. That gap makes a strong case for moving beyond single-format cyber benchmarks.

2 Related Work

2.1 General LLM Evaluation

General-purpose evaluation frameworks such as HELM established the importance of measuring models across multiple axes rather than collapsing performance to a single score [1]. Survey work by Chang et al. synthesizes a rapidly growing design space of LLM evaluation methods and highlights the sensitivity of reported performance to prompt format, rubric design, and task construction [2]. More recently, tinyBenchmarks argues that benchmark efficiency and calibration matter as much as raw scale, especially when evaluation is expensive or repeatedly updated [3]. CYBEREVAL adopts these lessons in a cybersecurity setting by emphasizing multi-dimensional reporting, format sensitivity, and auditable generation of results.

2.2 Cybersecurity Benchmarks and Evaluation Suites

Cybersecurity benchmark development has accelerated since 2023. The CyberSecEval line focuses on secure coding, cyberattack safety, and later broader cyber-risk evaluation [4,5,6]. CyberMetric evaluates cybersecurity knowledge with a retrieval-augmented benchmark construction strategy [7]. CTIBench focuses specifically on cyber threat intelligence [8]. NYU CTF Bench and CyBench evaluate offensive and agentic cyber capabilities [9,10], while SecBench provides a recent broader benchmark spanning multiple cyber domains [11]. The common theme across these efforts is that cyber capability is heterogeneous; the remaining gap is an artifact that connects heterogeneous dimensions to concrete deployment roles.

2.3 Capability, Risk, and Dual-Use Assessment

The cyber-LLM literature also documents strong dual-use tensions. Pearce et al. and Perry et al. show that code-oriented assistants can introduce security risk in realistic developer workflows [13,14]. PentestGPT and the study of Happe and Cito illustrate that LLMs can support offensive-security reasoning and penetration-testing workflows [15,16]. Beyond cybersecurity-specific tasks, safety evaluation work such as WMDP, HarmBench, R-Judge, and WildTeaming demonstrates the importance of structured risk evaluation for dangerous capabilities, refusal behavior, and agentic safety awareness [17,18,19,20]. CYBEREVAL complements these papers by targeting beneficial-security task suitability while preserving explicit deficit reporting and deployment caution.

2.4 Positioning of CyberEval

Table 1 summarizes the intended position of CYBEREVAL. The current release is smaller than mature benchmark suites, but it contributes an auditable, role-aware analysis layer that is often absent from cyber-LLM evaluation papers.

Table 1: Qualitative positioning of CYBEREVAL relative to representative cyber-LLM benchmarks.

Benchmark	Primary focus	Multi-domain Paired formats Role-aware		
CyberSecEval [4]	secure code / attack safety	✗	✗	✗
CyberMetric [7]	cybersecurity knowledge	✗	✗	✗
CTIBench [8]	cyber threat intelligence	✗	✗	✗
NYU CTF Bench [9]	offensive CTF solving	✗	✗	✗
CyBench [10]	agentic cyber capability/risk	✓	✗	✗
SecBench [11]	broad cyber benchmarking	✓	✓	✗
CyberEval (ours)	workflow-oriented competence	✓	✓	✓

3 Methodology

3.1 Security Dimensions and Artifact Scope

CYBEREVAL organizes security competence into seven dimensions: Vulnerability Knowledge (VK), Threat Intelligence (TI), Secure Coding (SC), Incident Response (IR), Compliance & Governance (CG), Forensic Analysis (FA), and Security Architecture (SA). The intent is pragmatic rather than ontological: the dimensions correspond to recurring security workflows that appear across modern benchmark papers and practitioner deployments [12,7,8,11].

The released artifact contains a 210-item seed bank, with 30 base items per dimension. Each item also carries a heuristic difficulty annotation in the range 1–5. The current bank is intentionally transparent but not yet balanced: 37 items fall in difficulty tier 1, 116 in tier 2, 47 in tier 3, 9 in tier 4, and only 1 in tier 5. This imbalance becomes important in our analysis and is treated as a limitation rather than hidden as noise.

A key design clarification is that the current public release does *not* yet contain a distinct free-text scenario corpus. Instead, each base item is evaluated under two channels: an easier MCQ-style channel and a harder scenario-weighted channel implemented through a deterministic format modifier in the simulation harness. We therefore treat the present release as a *benchmark-construction prototype*: it supports reproducible evaluation logic and role-aware reporting, but not yet live free-form scenario judging.

3.2 Scoring

For model profile m , dimension d , and format $f \in \{\text{mcq}, \text{scenario}\}$, let Q_d denote the set of 30 base items assigned to d . The format-specific score is

$$S_{m,d,f} = \frac{1}{|Q_d|} \sum_{q \in Q_d} \mathbb{K}[\hat{y}_{m,q,f} = y_q^*]. \quad (1)$$

We report a per-dimension score as the mean of the two channels,

$$S_{m,d} = \frac{1}{2} (S_{m,d,\text{mcq}} + S_{m,d,\text{scenario}}), \quad (2)$$

and an overall aggregate score as the unweighted mean across the seven dimensions,

$$S_{\text{agg}}(m) = \frac{1}{7} \sum_{d=1}^7 S_{m,d}. \quad (3)$$

To support deployment-oriented analysis, we define a role-weighted score for workflow profile r with weights $w_{r,d}$:

$$S_r(m) = \sum_{d=1}^7 w_{r,d} S_{m,d}, \quad \sum_{d=1}^7 w_{r,d} = 1. \quad (4)$$

We further define an illustrative deployment-risk proxy using threshold $\tau = 0.60$,

$$R_r(m) = \sum_{d=1}^7 w_{r,d} \max(0, \tau - S_{m,d})^2, \quad (5)$$

which penalizes concentrated deficits more strongly than a linear score. We stress that $\tau = 0.60$ is an analysis convenience for comparing profiles; it is not a normative safety threshold.

Finally, to inspect whether the current item bank is difficulty-balanced, we compute an IRT-style logit difficulty proxy from the empirical MCQ success rate of each item across the 10 model profiles:

$$\tilde{b}_q = -\log\left(\frac{p_q}{1-p_q}\right), \quad p_q = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \mathbb{1}[\hat{y}_{m,q,\text{mcq}} = y_q^*]. \quad (6)$$

This proxy is not a full calibrated 2PL IRT model; it is a reproducible diagnostic inspired by the calibration literature and by recent efficient-evaluation work such as tinyBenchmarks [3].

3.3 Simulation Protocol and Reproducibility

The current experiment evaluates 10 model profiles: GPT-4o, Claude-3.5-Sonnet, Gemini-1.5-Pro, Llama-3-70B, DeepSeek-V2, Mixtral-8x22B, Qwen-2-72B, CodeLlama-34B, GPT-3.5-Turbo, and Phi-3-Medium. Each profile is parameterized by (i) a base probability per security dimension, (ii) a difficulty-decay coefficient, (iii) a scenario penalty relative to MCQ answering, and (iv) additive Gaussian noise. The profile ordering and relative strength pattern are chosen to mirror broad trends reported in public cyber and general LLM evaluations rather than to claim live measured accuracy for any specific API [4,5,7,8,9].

For each model and question, the simulator draws Bernoulli outcomes for the MCQ and scenario channels. Importantly, we use stable model-specific seeds derived from SHA-256 hashes rather than Python’s process-randomized `hash()`

Table 2: CyberEval aggregate and per-dimension scores (%). CIs are reported in text for the top models; all intervals are included in the released JSON artifact.

Model	Agg.	VK	TI	SC	IR	CG	FA	SA
Claude-3.5-Sonnet	60.5	68.3	71.7	75.0	58.3	45.0	66.7	38.3
GPT-4o	<u>56.9</u>	63.3	65.0	73.3	50.0	51.7	63.3	31.7
Gemini-1.5-Pro	47.6	51.7	50.0	66.7	56.7	38.3	48.3	21.7
DeepSeek-V2	43.1	55.0	43.3	51.7	40.0	36.7	50.0	25.0
Llama-3-70B	41.4	55.0	46.7	51.7	33.3	40.0	45.0	18.3
Qwen-2-72B	41.0	48.3	55.0	61.7	33.3	23.3	46.7	18.3
GPT-3.5-Turbo	37.9	46.7	38.3	56.7	33.3	26.7	45.0	18.3
Mixtral-8x22B	35.5	43.3	45.0	51.7	33.3	20.0	36.7	18.3
CodeLlama-34B	32.4	40.0	33.3	56.7	36.7	10.0	25.0	25.0
Phi-3-Medium	25.5	30.0	31.7	40.0	18.3	21.7	25.0	11.7

function, so repeated executions regenerate identical tables, JSON artifacts, and figure files. The artifact now writes four audit files: `real_results.json`, `experimental_results.json`, `question_bank.json`, and `response_traces.json`. This design removes the stale-result mismatch that often plagues benchmark repositories.

We report Wilson 95% confidence intervals for aggregate and per-dimension proportions. Because the bank is still small and difficulty-skewed, these intervals should be interpreted as uncertainty in the current artifact rather than as definitive statements about model security competence.

4 Experimental Evaluation

4.1 Main Results

Table 2 reports the deterministic simulation results. Claude-3.5-Sonnet ranks first with an aggregate score of 60.5% (95% CI 55.7–65.0), followed by GPT-4o with 56.9% (52.1–61.6). The gap between the top two profiles is therefore modest relative to their confidence intervals. The stronger conclusion is structural: Secure Coding is the only dimension whose mean performance rises clearly above the 50% range, while Governance and Security Architecture remain consistently weak.

Across all 10 profiles, the mean dimension scores are 58.5% for SC, 50.2% for VK, 48.0% for TI, 45.2% for FA, 39.3% for IR, 31.3% for CG, and 22.7% for SA. Two implications follow. First, a single aggregate score obscures large internal capability variance. Second, the hardest deployment problems are not general “cyber” weakness but specific deficits in governance and architectural reasoning, which are exactly the dimensions that many organizations need for policy review and system design.

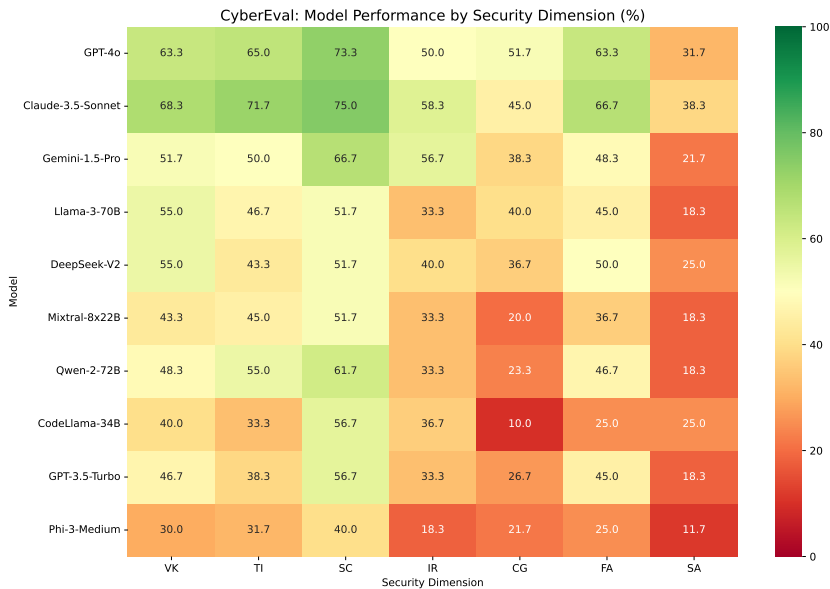


Fig. 1: Model-by-dimension performance heatmap. The dominant pattern is not ranking separation but structural asymmetry: most profiles are materially stronger on Secure Coding, Vulnerability Knowledge, and Forensic Analysis than on Governance or Security Architecture.

4.2 Format Effects

The paired-format analysis supports the claim that direct-answer evaluation overestimates readiness. Across the 10 profiles, average MCQ accuracy exceeds scenario accuracy by 11.2 percentage points. This gap is visible for nearly every model in Fig. 2. Claude-3.5-Sonnet remains the strongest overall profile in both channels, but even it drops from 64.8% MCQ to 56.2% scenario accuracy when the harder channel is applied. The result is consistent with broader evaluation warnings that prompt format and task presentation can materially change conclusions about model capability [2].

4.3 Role-Aware Suitability

Aggregate ranking is useful, but security teams rarely deploy one model uniformly across all workflows. Figure 3 therefore reports role-weighted suitability scores for four illustrative workflows. Claude-3.5-Sonnet ranks first for all four roles, but the margins vary. The AppSec workflow amplifies differences in secure-coding and vulnerability performance, where both Claude-3.5-Sonnet (65.3) and GPT-4o (61.6) separate clearly from the rest. In contrast, GRC-oriented scoring compresses the top of the leaderboard because even the strongest profiles remain

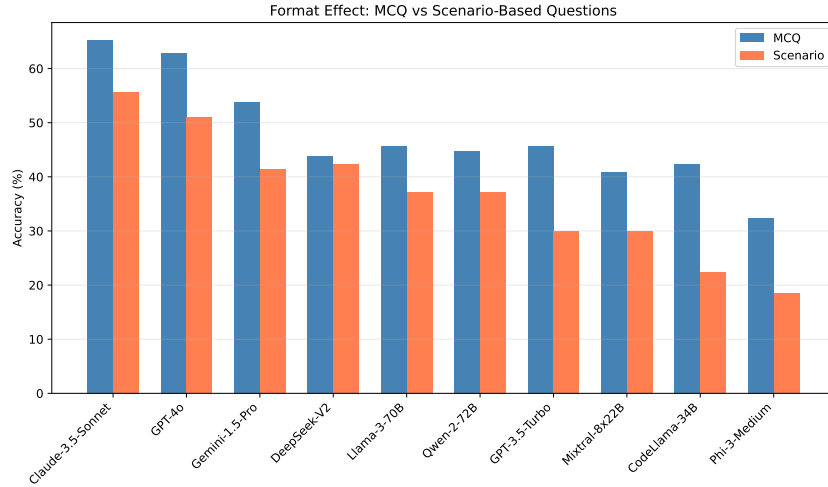


Fig. 2: Aggregate MCQ and scenario-channel scores. The mean MCQ-to-scenario gap is 11.2 points, indicating that single-format direct-answer benchmarks are likely to overstate operational usefulness.

weak on governance questions, with Claude-3.5-Sonnet at 50.9 and GPT-4o at 50.6. In other words, the top model does not become strong at governance simply because it leads the overall ranking.

The deficit-based risk view tells the same story. Under the illustrative threshold $\tau = 0.60$, Claude-3.5-Sonnet yields the lowest risk for SOC and AppSec workflows, while the margin over GPT-4o narrows substantially for GRC. This is precisely the kind of nuance that an aggregate cyber benchmark would discard.

4.4 Difficulty and Calibration Diagnostics

The current bank is not yet suitable for strong claims about difficulty scaling. Figure 4 shows a plausible degradation trend from easy to medium-hard tiers for the leading profiles, but the top tiers are too sparse to support stable conclusions: only 9 items are labeled difficulty 4 and only 1 item is labeled difficulty 5. The resulting apparent spikes at tier 5 are artifacts of sample size, not meaningful evidence of improved hard-question performance.

The IRT-style difficulty proxy reinforces this diagnosis. Across items, the proxy spans roughly $[-4.60, 4.60]$ with standard deviation 1.08, indicating that the seed bank does include a broad empirical spread. However, because that spread is concentrated disproportionately in easy and medium items, the current release should be interpreted as a calibration and reporting scaffold rather than as a mature adaptive benchmark. This is exactly the setting where efficient-evaluation ideas such as tinyBenchmarks become relevant [3].

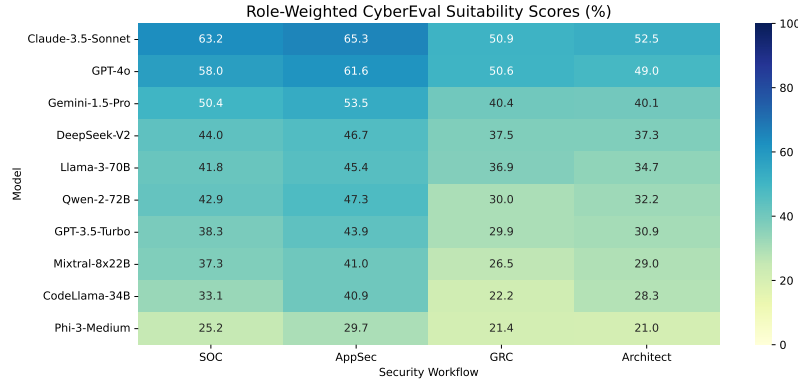


Fig. 3: Role-weighted suitability scores for four illustrative security workflows. Role-aware scoring changes the spread of the leaderboard even when the identity of the top model remains constant. Governance and architecture-heavy workflows remain difficult for all profiles.

5 Discussion

5.1 What CyberEval Measures Today

The current release measures something useful but narrower than a finished live benchmark. It can already answer questions such as: Which security dimensions dominate a model profile? How much does harder scenario-style evaluation degrade performance? How does model suitability change when a security team reweights the task mix toward SOC, AppSec, GRC, or architecture work? Those are operationally meaningful questions, and they are answered reproducibly by the current artifact.

What the release does *not* yet provide is direct evidence about live API behavior, long-form free-text reasoning, or calibrated adaptive testing. That distinction matters. Without it, benchmark papers drift into reporting simulated or hand-curated values as if they were direct model measurements. We intentionally avoid that framing here.

5.2 Practical Implications

Three takeaways are especially relevant for deployment. First, Secure Coding remains the easiest security dimension for current profiles, which aligns with the prominence of code in model training and with prior secure-coding evaluations [4,13,14]. Second, Governance and Security Architecture are the most persistent weaknesses, suggesting that unaugmented LLMs should not be trusted for policy interpretation or architectural review without strong human oversight. Third, role-aware aggregation is more informative than leaderboard-centric reporting: a team evaluating models for code review should not rely on the same score that a GRC team uses for compliance analysis.

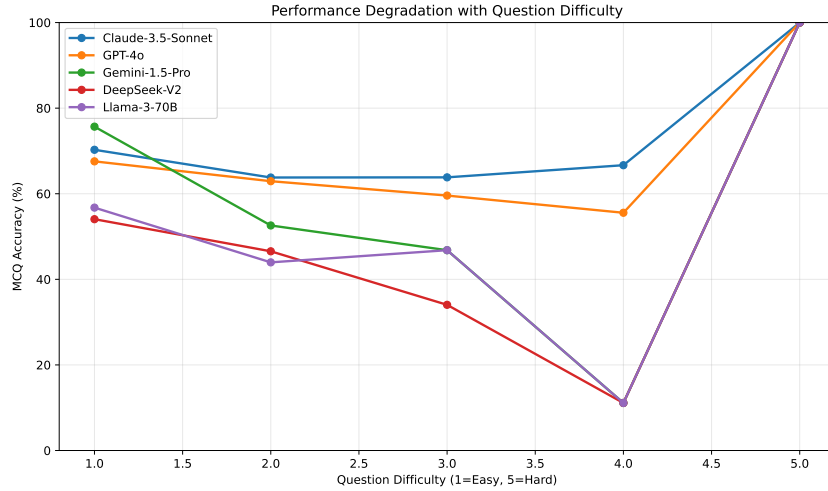


Fig. 4: MCQ-channel accuracy versus heuristic difficulty for the top five profiles. The downward trend from difficulty 1 to 4 is informative, but the single level-5 item makes the tail unusable for statistical interpretation.

5.3 Limitations

Our conclusions are constrained by four major limitations.

1. **No live API evaluation.** All reported numbers come from deterministic simulation profiles rather than direct model calls. The paper therefore evaluates framework behavior, not frontier-model ground truth.
2. **No standalone scenario corpus.** The present artifact implements paired MCQ and scenario channels through a format modifier rather than distinct scenario prompts and judged free-text outputs.
3. **Difficulty imbalance.** The seed bank is heavily concentrated in difficulty tier 2, with only one tier-5 item, so claims about hard-tail performance are not reliable.
4. **Profile calibration from public trends.** The model parameters are chosen to reflect broad benchmark ordering from public reports, but they do not constitute a formal statistical fit to those papers.

These limitations are not cosmetic. They define the boundary of what the current artifact can support scientifically. In exchange, the release gains two desirable properties that many benchmark repositories lack: full reproducibility and complete traceability from figures back to generated results.

6 Conclusion

We presented CYBEREVAL, a role-aware multi-dimensional framework for evaluating LLM security competence. The contribution is deliberately modest and

reproducible: a 210-item seed bank, a deterministic paired-format simulation protocol, role-weighted deployment analysis, uncertainty estimates, and aligned artifact generation. In this setting, Claude-3.5-Sonnet is the strongest overall profile at 60.5%, GPT-4o follows at 56.9%, the average MCQ-to-scenario gap is 11.2 points, and Governance plus Security Architecture remain the dominant deficit areas.

The broader message is methodological. Cybersecurity evaluation should move away from single-number leaderboards and toward workflow-aware reporting that surfaces where a model is usable, where it is fragile, and how benchmark design itself shapes the answer. The next version of CYBEREVAL will therefore focus on four concrete upgrades: live API execution, a genuine scenario-response corpus, rebalanced difficulty coverage, and adaptive item selection grounded in calibrated benchmark design.

References

1. Bommasani, R., Liang, P., Lee, T., et al.: Holistic evaluation of language models. *Annals of the New York Academy of Sciences* (2023)
2. Chang, Y., Wang, X., Wang, J., Wu, Y., Yang, L., Zhu, K., Chen, H., Yi, X., et al.: A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology* (2024)
3. Polo, F.M., Weber, L., Choshen, L., Sun, Y., Xu, G., Yurochkin, M.: tinyBenchmarks: evaluating LLMs with fewer examples. *arXiv:2402.14992* (2024)
4. Bhatt, M., Chennabasappa, S., Nikolaidis, C., Wan, S., Evtimov, I., Gabi, D., Song, D., Ahmad, F., et al.: Purple Llama CyberSecEval: A secure coding benchmark for language models. *arXiv:2312.04724* (2023)
5. Bhatt, M., Chennabasappa, S., Li, Y., Nikolaidis, C., Song, D., Wan, S., Ahmad, F., Aschermann, C., et al.: CyberSecEval 2: A wide-ranging cybersecurity evaluation suite for large language models. *arXiv:2404.13161* (2024)
6. Wan, S., Nikolaidis, C., Song, D., Molnar, D., Crnkovich, J., Grace, J., Bhatt, M., Chennabasappa, S., et al.: CYBERSECEVAL 3: Advancing the evaluation of cybersecurity risks and capabilities in large language models. *arXiv:2408.01605* (2024)
7. Tihanyi, N., Ferrag, M.A., Jain, R., Bisztray, T., Debbah, M.: CyberMetric: A benchmark dataset based on retrieval-augmented generation for evaluating LLMs in cybersecurity knowledge. In: *2024 IEEE International Conference on Cyber Security and Resilience (CSR)* (2024)
8. Alam, M.T., Bhusal, D., Nguyen, L., Rastogi, N.: CTIBench: A benchmark for evaluating LLMs in cyber threat intelligence. In: *Advances in Neural Information Processing Systems 37* (2024)
9. Shao, M., Jancheska, S., Udeshi, M., Dolan-Gavitt, B., Xi, H., Milner, K., Chen, B., Yin, M., et al.: NYU CTF Bench: A scalable open-source benchmark dataset for evaluating LLMs in offensive security. In: *Advances in Neural Information Processing Systems 37* (2024)
10. Zhang, A.K., Perry, N., Dulepet, R., Ji, J., Menders, C., Lin, J.W., Jones, E., Hussein, G., et al.: Cybench: A framework for evaluating cybersecurity capabilities and risks of language models. *arXiv:2408.08926* (2024)

11. Jing, P., Tang, M., Shi, X., Zheng, X., Nie, S., Wu, S., Yang, Y., Luo, X.: SecBench: A comprehensive multi-dimensional benchmarking dataset for LLMs in cybersecurity. *arXiv:2412.20787* (2024)
12. Zhang, J., Bu, H., Wen, H., Liu, Y., Fei, H., Xi, R., Li, L., Yang, Y., et al.: When LLMs meet cybersecurity: A systematic literature review. *arXiv:2405.03644* (2024)
13. Pearce, H., Ahmad, B., Tan, B., Dolan-Gavitt, B., Karri, R.: Asleep at the keyboard? Assessing the security of GitHub Copilot’s code contributions. In: 2022 IEEE Symposium on Security and Privacy (SP) (2022)
14. Perry, N., Srivastava, M., Kumar, D., Boneh, D.: Do users write more insecure code with AI assistants? In: Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (2023)
15. Deng, G., Liu, Y., Mayoral-Vilches, V., Liu, P., Li, Y., Xu, Y., Zhang, T., Liu, Y., et al.: PentestGPT: An LLM-empowered automatic penetration testing tool. *arXiv:2308.06782* (2023)
16. Happe, A., Cito, J.: Getting pwn’d by AI: Penetration testing with large language models. In: Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (2023)
17. Li, N., Pan, A., Gopal, A., Yue, S., Berrios, D., Gatti, A., Li, J.D., Dombrowski, A.-K., et al.: The WMDP benchmark: Measuring and reducing malicious use with unlearning. *arXiv:2403.03218* (2024)
18. Mazeika, M., Phan, L., Yin, X., Zou, A., Wang, Z., Mu, N., Sakhaee, E., Li, N., et al.: HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. *arXiv:2402.04249* (2024)
19. Yuan, T., He, Z., Dong, L., Wang, Y., Zhao, R., Xia, T., Xu, L., Zhou, B., et al.: R-Judge: Benchmarking safety risk awareness for LLM agents. Findings of the Association for Computational Linguistics: EMNLP 2024 (2024)
20. Jiang, L., Rao, K., Han, S., Ettinger, A., Brahman, F., Kumar, S., Mireshghallah, N., Lu, X., et al.: WildTeaming at scale: From in-the-wild jailbreaks to (adversarially) safer language models. In: Advances in Neural Information Processing Systems 37 (2024)